

Denizhan Kara

kara4@illinois.edu | denizhankara.github.io | github.com/denizhankara | linkedin.com/in/denizhankara

EDUCATION

University of Illinois at Urbana-Champaign

Champaign, IL

Ph.D. in Computer Science - Siebel School of Computing and Data Science

Aug. 2022 – Expected May 2026

– **Focus:** Foundation Models, Multimodal Learning, Physics-Informed AI

University of Illinois at Urbana-Champaign

Champaign, IL

M.S. in Computer Science - Siebel School of Computing and Data Science

Aug. 2020 – Dec. 2021

Koç University

Istanbul, Turkey

B.Sc. in Electrical and Electronics Engineering & Physics (Double Major)

Sep. 2012 – Jun. 2017

– Full-Merit Scholarship & Turkish Prime Ministry Scholarship (top 100 in National Exam, 2M+ candidates)

WORK EXPERIENCE

Software Engineer Intern - Driver Understanding ML

May 2025 – Aug. 2025

Waymo

Mountain View, CA

- Deployed Gemini LLM automation framework serving millions of daily AV events, achieving 91% recall on Traffic Events Triage covering 35% of queries.
- Built production few-shot inference pipeline, improving complex judgment recall by 9 points.
- Developed prompt engineering guides and automation pipelines adopted by 3 teams, enabling up to 90% task recall.
- Implemented dynamic context-aware prompting using real-time sensory data for adaptive automation.

Research Scientist Intern - Multimodal Foundation Models R&D

Jun. 2024 – Sep. 2024

US Army Research Laboratory (DEVCOM ARL)

Adelphi Laboratory Center, MD

- Developed sub-10M parameter multimodal foundation models for vehicle classification, achieving 12% accuracy improvement with real-time edge deployment.
- Integrated physical decay models into pretraining with novel loss functions, improving environmental robustness by 10% on low-cost edge devices.
- Deployed real-time inference pipeline with sub-second latency for distributed autonomous sensing in resource-constrained environments.

Senior Consultant - ML Engineer

Jan. 2020 – Aug. 2022

Prometeia SpA

Istanbul, Turkey

- Architected AI propensity scoring framework processing 3M+ transactions using time series embeddings and attention mechanisms for customer interest prediction.
- Developed deep learning credit default system with novel temporal features, improving recall by 25%.
- Enhanced Allianz Insurance damage assessment system with segmentation models and augmentation pipeline, improving F1-score by 6%.

Software Design Engineer

Jul. 2017 – Dec. 2019

Turkish Aerospace - Autopilot Systems Division

Ankara, Turkey

- Developed signal processing libraries for autopilot control systems, reducing processing delay by up to 20%.
- Led interpretation of electromagnetic and vibrational noise in sensor data, developing filtering solutions compliant with control algorithms, and created a sensor emulator framework for software-in-the-loop simulations.

RESEARCH EXPERIENCE

Graduate Researcher - Foundation Models

Aug. 2022 – Present

Physics-Informed AI for Distributed IoT Systems

CyPhy Research Group at UIUC

- Developed foundation models integrating domain knowledge and physical laws with state-of-the-art self-supervised learning techniques for IoT Foundation Models.
- Aiming to build efficient, lightweight, and explainable AI systems for IoT applications deployable on edge devices.
- Collaborated with a multidisciplinary team, resulting in publications in top-tier conferences such as ACM SenSys, WWW, and NeurIPS.

Graduate Researcher - Adversarial ML & Security

Aug. 2020 – May 2022

Autonomous Vehicle & UAV Security Research

Systems Security Research Group at UIUC

- Designed OVERTON, a misbehavior detection framework for V2X networks using temporal anomaly detection and ML-based trust mechanisms to defend against adversarial attacks.
- Developed adversarial attack framework for UAV swarms using sensor-spoofing strategies to manipulate IMU readings, improving defensive security protocols for autonomous systems.

SELECTED PUBLICATIONS

- [1] **Kara, D.**, Kimura, T., Li, J., He, B., Chen, Y., Hu, Y., Zhao, H., Liu, S., & Abdelzaher, T. (2025). *AdaTS: Learning Adaptive Time Series Representations via Dynamic Soft Contrasts*. Advances in Neural Information Processing Systems (NeurIPS 2025).
- [2] **Kara, D.**, Kimura, T., Sun, D., Li, J., Chen, Y., Hu, Y., Zhao, H., Bhattacharyya, J., & Abdelzaher, T. (2025). *DiffPhys: Differential Physics Augmentations for Enhanced Representations*. 34th International Conference on Computer Communications and Networks (ICCCN 2025).
- [3] Hu, C., Chen, Y., **Kara, D.**, Liu, S., Abdelzaher, T., Wu, F., & Chen, G. (2025). *OpenMAE: Efficient Masked Autoencoder for Vibration Sensing with Open-Domain Data Enrichment*. Proceedings of the ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies (ACM IMWUT), also presented at UbiComp 2025, Espoo, Finland.
- [4] Kimura, T., Li, X., Hanna, O., Chen, Y., Chen, Y., **Kara, D.**, Wang, T., Li, J., Ouyang, X., Liu, S., Srivastava, M., Diggavi, S., & Abdelzaher, T. (2025). *InfoMAE: Pairing-Efficient Cross-Modal Alignment with Informational Masked Autoencoders for IoT Signals*. ACM TheWebConference (WWW 2025), Sydney
- [5] **Kara, D.**, Kimura, T., Chen, Y., Li, J., Wang, R., Chen, Y., Kaplan, L., Bhattacharyya, J., & Abdelzaher, T. (2024). *PhyMask: An Adaptive Masking Paradigm for Efficient Self-Supervised Learning in IoT*. 22nd ACM Conference on Embedded Networked Sensor Systems (SenSys 2024).
- [6] **Kara, D.**, Kimura, T., Liu, S., Li, J., Liu, D., Wang, T., Wang, R., & Abdelzaher, T. (2024). *FreqMAE: Frequency-Aware Masked Autoencoder for Multi-Modal IoT Sensing*. ACM Web Conference (WWW 2024)
- [7] Wang, T., Yang, Q., Wang, R., Sun, D., Li, J., Chen, Y., Hu, Y., Yang, C., Kimura, T., **Kara, D.**, & Abdelzaher, T. (2024). *Fine-grained Control of Generative Data Augmentation in IoT Sensing*. Advances in Neural Information Processing Systems (NeurIPS 2024).
- [8] **Kara, D.**, Kim, K. H., Mohan, S., Hasan, M., Shimizu, T., & Lu, H. *OVERTON: A Misbehavior Detection and Trust Framework for Vehicular (V2X) Networks*. Under review at USENIX Security Symposium 2025.

Selected Conference Presentations & Abstracts

- [9] **Kara, D.**, Bhattacharyya, J., Goldman, G., & Abdelzaher, T. (2024). *Integrating Physical Principles with Self-Supervised Learning for Robust Vehicle Classification in Diverse Environments*. AGU Fall Meeting Abstracts 2024, IN43A-2256.
- [10] **Kara, D.**, Bhattacharyya, J., & Abdelzaher, T. (2025). *A Novel Physics-Guided Contrastive Learning Strategy for Seismic Signal Analysis*. 2025 Military Sensing Symposium Joint (BAMS and NSSDF) Conference.

Full publication list (30+ papers) available at denizhankara.github.io

TECHNICAL SKILLS

Foundation Models & LLMs: Large-scale pretraining, sequence modeling, Prompt engineering
Deep Learning: PyTorch, TensorFlow, Hugging Face Transformers, JAX, CUDA
Machine Learning: Time series, Multimodal learning, Self-supervised learning, Domain-aware learning
Programming: Python, C++, MATLAB, Distributed training, Model deployment
Infrastructure: AWS/GCP, Docker, Spark, Real-time inference